

Mahamat Nour Bachar MAHAMAT

Application for a PhD position
Deep Learning & Computer Vision

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Experience

Bioacoustic event detection and source separation using NMF

May – Oct. 2025

IMT Atlantique, Brest — MSc research internship

- Designed an **unsupervised** method for sound event detection: nonnegative matrix factorization (*NMF*) combined with temporal segmentation (*ruptures*), requiring no training phase; NMF components correspond to spectral patterns.
- Validated on BioDCASE 2025 data (task 2) and compared against the challenge's supervised baselines (YOLOv11, ResNet18). On ElephantIsland2013: F1 of 0.20 in 5 min on CPU, versus 0.41 for YOLOv11 trained on GPU.

Student jobs — Alliance Healthcare, Kuehne+Nagel, McDonald's (*alongside studies*)

2021 – 2025

Education

MSc in Mathematics and Applications, Data Science track — Université d'Angers, France

2023 – 2025

Main coursework ([full curriculum](#)):

- Statistics:** high-dimensional statistics, Bayesian inference, hypothesis testing, time series (Prophet, ARIMA, ...).
- Machine learning:** generative models, image processing (classification, segmentation, detection), NLP, PINN.
- Numerical methods:** optimization (gradient-based, constrained), signal processing (FFT, filtering, sampling).
- Software engineering:** Python (OOP), SQL, MLOps, big data.

BSc in Mathematics — Université d'Angers, France

2020 – 2023

PluriPASS (*formerly PACES*, first year of health science studies) — Université d'Angers, France

2019 – 2020

Biological sciences, biostatistics, scientific methodology.

Baccalauréat, série D (High school diploma, Mathematics & Biology) — Lycée La Fidélité, Chad

2018

National rank: 22nd/79,136

Scientific Communication

A. Marmoret, M. Mahamat, D. Cazau. “Unsupervised detection of bioacoustic events by nonnegative matrix factorization”. *Oral presentation*, SERENADE Workshop, Brest, June 2026, [hal-05668390v1](#). A draft paper is in preparation.

Research Projects (Academic)

Generative diffusion models — MSc thesis, [Read the report](#)

Nov. 2024 – Mar. 2025

- Study of the theoretical foundations: stochastic differential equation formulation, time reversal and connection to score matching.
- Experimental comparison of diffusion / GAN / VAE on MNIST: evaluation of sample diversity (Mean Max Logit) and class coverage.

Approximate Bayesian Computation with random forests (ABC-RF) - First-year MSc project

Feb. – May 2024

- Reproduced the method of [Pudlo et al. \(2016\)](#) for model choice in likelihood-free Bayesian inference.
- Validation on simulated data, then application to inferring the evolutionary history of Asian rice (R, `abcrf`).

Projects

Biomedical imaging and signals

ECG signal denoising (personal project)

2026

- Added simulated physiological noise (Gaussian, electrode motion, muscle artifacts, ...) to ECG recordings.
- Designed and trained convolutional networks for denoising physiological signals.

Blood cell segmentation

Feb. 2025

- Compared a U-Net trained end-to-end against a pre-trained VGG16 encoder used via transfer learning.
- Data augmentation (rotations, elastic deformations, brightness variations, ...) applied to the transfer-learning model to mitigate the observed overfitting; Dice of 0.97 (U-Net) versus 0.90 (*fine-tuned* VGG16).

Disease/healthy classification (breast cancer, GSE116)

Dec. 2024 – Jan. 2025

- Preprocessing and normalization of high-dimensional genomic data ($n \ll p$).
- Feature selection by ANOVA; comparison of supervised classifiers for disease/healthy discrimination.

Object detection - Implemented an object detection pipeline based on <i>YOLO11</i>, with <i>fine-tuning</i> on the MinneApple dataset. - Compared the performance of <i>DINO</i> (foundation model) against <i>YOLO</i>, evaluated by mAP/IoU.	Feb. 2025
Natural language processing: text classification and generation - Fine-tuning of BERT variants on two tasks: news classification (<i>fake news</i>) and headline generation. - Evaluation with supervised metrics (F1-score) and qualitative analysis of the generated text.	Feb. 2025
Molecule–protein affinity prediction (ENS Paris-Saclay challenge) - Molecular encoding (SMILES → ECFP <i>Morgan fingerprints</i>, 2048 bits) with RDKit; regression with XGBoost. - Prediction of the half-maximal inhibitory concentration (pIC50) on a set of 4,400 molecules; evaluation by cross-validation.	Dec. 2024 – Jan. 2025
Generative models: GAN and VAE Implementation and training, in preparation for the MSc thesis on diffusion models.	2024
Reinforcement learning: DQN and PPO Agents trained on <i>GridWorld</i> , <i>CartPole</i> and <i>LunarLander</i> ; study of the contribution of <i>experience replay</i> (prioritized or not) and of the <i>target network</i> ; PPO with adaptive KL penalty.	2024

Statistics and data engineering

Determinants of life expectancy worldwide (WHO data) - Study covering 193 countries (2000–2015) to identify factors influencing longevity. - Preprocessing and application of the following models: OLS, mixed effects, instrumental variables.	Mar. 2025 – Jun. 2025
Parameter estimation in dependent stationary series - Study of the bias and asymptotic variance of estimators (mean, variance, quantiles) under dependence. - Application of mixing conditions (Bradley, Rio E.) to assess the reliability of the estimated parameters.	Dec. 2024 – Jan. 2025
Database and web interface — Enron dataset - Data extraction using regular expressions (regex), data structuring and population of a PostgreSQL database. - Development of a web application with Django + HTML for interactive visualization.	Jan. 2024 – May 2024

Further details on my [Git](#) and [portfolio](#).

Skills

Languages: Python, R, SQL, Unix/bash; C++ (basics). Scientific computing: NumPy, SciPy, scikit-learn, PyTorch; parallelization (multiprocessing, asyncio). Statistics: Bayesian inference, Monte Carlo, high dimension ($n \ll p$), time series, estimation under dependence. Machine learning: regression, classification, clustering, dimensionality reduction, cross-validation. Deep learning: CNN, autoencoders, Transformers.	Signal & image: Fourier analysis, filtering, sampling; segmentation, detection, denoising. NLP: tokenization, <i>embeddings</i>, model <i>fine-tuning</i>. Data: PostgreSQL, SQLite, MongoDB, regex, <i>web scraping</i>; PySpark (basics). MLOps & tools: Git, Docker, conda, venv, Django/Flask, L^AT_EX. Visualization: Matplotlib, Seaborn, ggplot2, Power BI.
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Research skills

- Reading and reproducing scientific papers.
- Writing reports and presenting results.
- Working independently on unfamiliar topics.

Languages

- English:** fluent reading and writing
- French:** bilingual
- Arabic:** bilingual

References

Dr. Axel Marmoret — Associate Professor, IMT Atlantique Supervisor of my MSc internship. axel.marmoret@imt-atlantique.fr Prof. Gilles Stupfler — Professor, Université d’Angers Statistics lecturer; head of admissions for the MSc programme. gilles.stupfler@univ-angers.fr	Prof. David Rousseau — Professor, Université d’Angers Image processing lecturer & Head of ImHorPhen. david.rousseau@univ-angers.fr Prof. Sylvain Lamprier — Professor, Université d’Angers Lecturer in reinforcement learning, generative models and NLP. sylvain.lamprier@univ-angers.fr
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Miscellaneous

- Involvement:** supporting international students newly arrived in France (integration, administrative procedures).
- Interests:** running, board games, science communication and history.